



ECOLOGICALLY SAFE, ENVIRONMENTALLY FRIENDLY, EFFECTIVE AFFORDABLE SLOT DISCHARGE METHOD. MAXIMIZE PRODUCTION EXTEND WELL LIFE

1

OIL AND GAS PRODUCT INTENSIFICATION AND RETRIEVAL TECHNOLOGY

MAXXWELL PRODUCTION INC.

OIL WELL SLOTTING PROFESSIONAL SERVICE

72 Beechbrooke way, Aurora, ON Canada L4G 6N7 Tel: 1-800-696-5721, (716) 359-2559, (647) 367-1810, Fax: (845) 746-4340



maxxwell production
well slotting service

MAXXWELL PRODUCTION INC.

MAXIMIZE PRODUCTION
EXTEND WELL LIFE

**SLOTTING PERFORATION CUT PROGRAM
REDMAN D Well Number: 1 142090023639
GRANITE WASH (MONTAGUE, TEXAS)**

**ECOLOGICALLY SAFE,
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FRIENDLY, EFFECTIVE
AFFORDABLE METHOD**

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(647) 367-1810
Fax: (845) 746-4340
(647) 724-4308

E-mail: internet@maxxwell.ca
internet@maxxwell.net

Web: <http://www.maxxwell.ca>
<http://www.maxxwell.net>



Maxxwell Production Inc.



Stanford University

(Department of Geophysics)

"Abrasive hydro-slotting perforation is simple, but very effective method of well production enhancement which incorporates modern concepts including the interaction between reservoir mechanics and the physical permeability of reservoir rocks.

Based on the available information, including theoretical calculation, laboratory experiments, and field test results, we believe that this method may find effective application in the oil and mining industries."

Amos Nur

Prof. of Geophysics and Director,
Stanford Rock Physics and Borehole
Geophysics Project
Stanford University

Lev Vernik

Research Associate
Stanford Rock Physics and Borehole
Geophysics Project
Stanford University





January 29, 2014

TX-001

SLOT DISCHARGE OF PRODUCTIVE LAYER WITH HYDRO-SLOTTING PERFORATION

OIL WELL **REDMAN D** Well Number: **1** API **142090023639**
WES POTEET, GRANITE WASH FORMATION, MONTAGUE, TEXAS

Date: January 29, 2014

By: Anatoli Nikouline CEO, SPE, C.E.T. SASTT
Slotting perforation technology engineer
Maxxwell Production Inc.
72 Beechbrooke way,
Aurora, Ontario, Canada
L4G 6N7

For: Christian Weber,
President
ReduxEnergy,
963 Topsy Lane, Suite 306-261
Carson City, Nevada
United States 89705-8417

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OIL WELL **REDMAN D** Well Number: **1** API **142090023639** WES POTEET, GRANITE WASH FORMATION, MONTAGUE, TEXAS

WELL DATA

142090023639

County, State: MONTAGUE, TEXAS

Lease: REDMAN D Well Number: 1

Operator: ROCKING R DRLLG&PROD (028853)

Status: INACTIVE

Prod ID: 142090023639

Production Thru Date: Aug 31, 2012

Header Block

Data Source:	PI	Completion Date:	Aug 21, 1983
Lease Name:	REDMAN D (023639)	Total Depth:	2,959 FT
Operator:	ROCKING R DRILLING & PROD COMPANY	Upper Perforation:	2,670 FT
State:	TEXAS	Lower Perforation:	2,685 FT
County:	MONTAGUE	Oil Gravity:	
District:	TEXAS DISTRICT 09	Gas Gravity:	
Primary API:	42337321050000	Last Production Date:	Aug 31, 2012
Regulatory:		Temp Gradient:	
First Production Date:	Sep 01, 1983	N Factor:	
Field:	WES POTEET	Prod Zone Code:	409GRWS
Reservoir Name:	GRANITE WASH 2670	GOR:	1000 CFB
Prod Zone:	GRANITE WASH/PENNSYLVANIAN	Play Type:	
Basin Name:	FT WORTH SYNCLINE (420)	Primary Product:	CRUDE OIL
Play Name:		Status:	INACTIVE
Gas Gatherer:		Data Source:	PI
Liquid Gatherer:	SENTI	Lat/Long Source:	IH
Location:	SEC: BLK: SRV: FORRESTER CHARLES ABS: 276		
Latitude/Longitude:	+33.79943 -97.48746		
Cum Data Source:	PI		
Cum Oil:	33,470 BBL	Cum Inj Liq:	
Cum Gas:	251 MCF	Cum Inj Gas:	
Cum Water:	100,636 BBL Since AUG 1984	Cum Inj Water:	





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Results of calculation the residual reserves:

(counted the reserves by method of accumulated production by the years)

OIL			GAS			WATER		
total reserve	gotten reserve	residual reserve	total reserve	gotten reserve	residual reserve	total reserve	gotten reserve	residual reserve
51008	33470	8501	332	252	12	142009	100636	14201

Welex
DIGITAL LOGGING SYSTEM

DUAL INDUCTION GUARD LOG

COMPANY **POTTEET OPERATING CO.**
WELL **REDMAN - D #1**
FIELD **MONTAGUE REGULAR**
COUNTY **MONTAGUE** STATE **TEXAS**
Location **330 F.W.L 330 FSL**
Charles Forester Survey
A - 276

Sec Two Rge
Perm. Datum **Ground Level** Elev **N/A**
Log Measured From **K.B. OR S** Ft Above Perm. Datum
Drill. Measured From **Kelly Bushing**

Date **7-31-83**
Run No **ONE**
Dep. **2960'**
Des. **2952'**
Bit **2950'**
Log **120'**
Gas **8 3/4" - 120'**
Casing **12 1/2"**
Bit Size **7 1/8"**
Type **DEPLETING MOB**

Dep. **9.3 134**
Fluid Loss **10 184 ml**
Sample **PIT**
W. Te **2.2 82**
W. Te **1.9 82**
W. Te **3.0 82**
Measured **1.8 107**
2 1/2 hrs
107' 8H
0701 WFT
By **M.M. Whitworth**
Checked By **MR. POTTEET**

REPRODUCED BY
Petroleum Information
Corporation
MIDLAND, TEXAS 79701

REFERENCE **N 8141Y**



12 COMPLETION RECORD

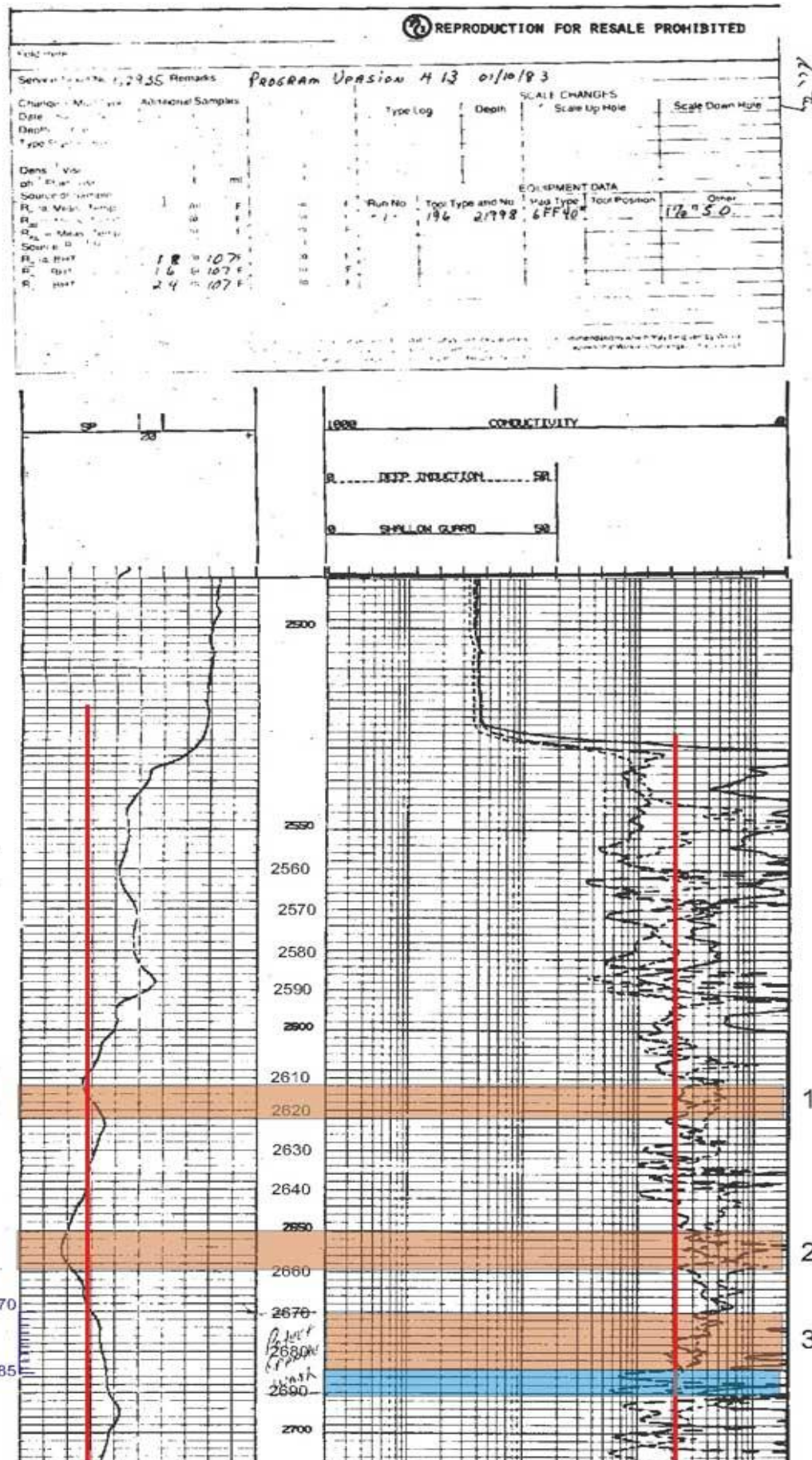


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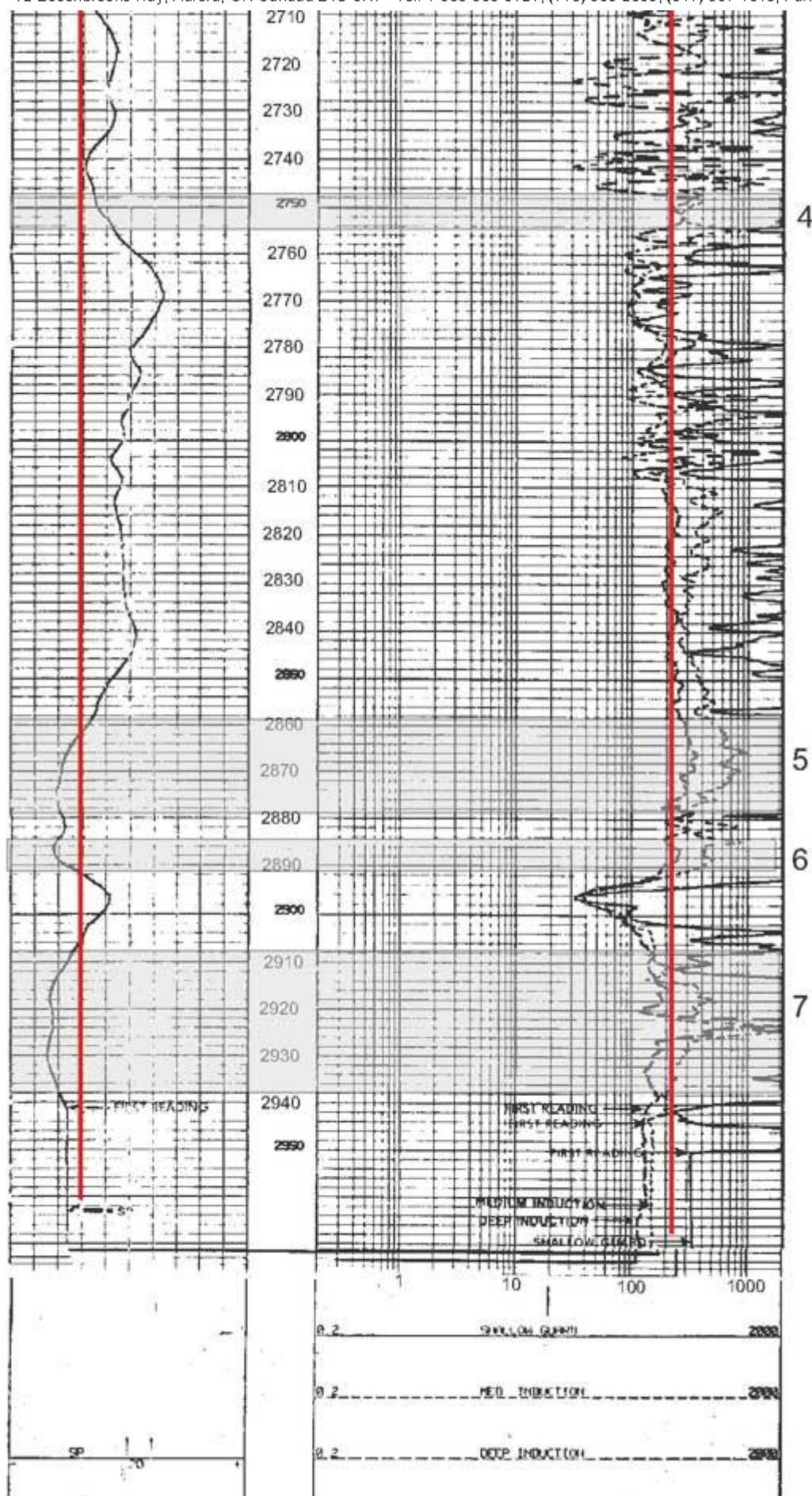
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**Analysis of the prospects of the formation of the well
Redman D1 API 142090023639****Baseline:**

Geophysical investigations (curve SS (SP), three curves of the resistance from probes on the different depth (DEEP, MED, SHALLOW), data on product selection from the well in the period from **1983** to **2009**.

According to the GIS data, in the presented well "Redmon D" formation is opened at a depth **2528** ft. The formation is provided by granitoid sandstones (possibly and limestone). In the incision of the well, by curves of the GIS, is marked collector. (Table 1, Figure 1) Resistance of oil saturated collectors is over 2000 ohmm, resistance of water saturated collectors is less than 900 ohmm. To evaluate the porosity in the collectors it is not possible due to lack of GIS datas for determining the porosity.

The interval of perforation is open the collector on the depths **2670-2685** ft, the lower part of which is flooded (**2684-2690** ft).

Table 1

#	roof (ft)	foot (ft)	thickness (ft)	saturation
1	2610	2622	12	oil
2	2648	2660	12	oil
3	2670	2684	14	oil + water
	summary		38	
4	2747	2755	8	possible oil
5	2858	2879	21	possible oil
6	2885	2892	7	possible oil
7	2908	2938	30	possible oil

From the results of the GIS interpretation follow:

1. Layers **1** and **2** are new objects of the development.
2. Layer **3** is located on the water-saturated formation, and re development of this layer can lead to an increase of water flow into the production well.
3. Layers **4, 5, 6, 7** according to the GIS data and geophysical investigations are analogues of the layers **1** and **2**, however to determine the real saturation is not possible.
4. Analysis of the residual reserves calculations results shows, that the minimum residual oil reserve does not exceed **30~40%**.



Table intervals recommended for slotting perforation:

# layer	roof (ft)	foot (ft)	thickness (ft)
1	2610	2611	1
	2611.5	2612.5	1
summary			2
2	2648	2649	1
	2649.5	2650.5	1
	2651	2652	1
	2652.5	2653.5	1
	2654	2655	1
	2655.5	2656.5	1
	2657	2658	1
	2658.5	2659.5	1
summary			8
3	2670	2671	1
	2671.5	2672.5	1
	2673	2674	1
	2674.5	2675.5	1
	2676	2677	1
	2677.5	2678.5	1
	2679	2680	1
summary			7

Total:

17 ft

Cutting intervals :

Interval 1	2677.50 - 2679.14 ft
Interval 2	2674.50 - 2676.14 ft
Interval 3	2671.50 - 2673.14 ft
Interval 4	2657.50 - 2659.14 ft
Interval 5	2654.50 - 2656.14 ft
Interval 6	2651.50 - 2653.14 ft
Interval 7	2648.50 - 2649.14 ft
Interval 8	2610.00 - 2611.64 ft
Total intervals (continuous slot):	12.8 ft
Total intervals (1.36 ft between slots):	19.92 ft
Total time :	~ 16 hours



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ENTERING VALUES FOR THE CALCULATION OF THE TECHNOLOGICAL PARAMETERS FOR THE SLOTTING PERFORATION

OD Casing	4.5"
ID Casing	4.0"
Drift Casing	4.0"
OD Tubing	2.38"
ID Tubing	1.7"
Drift Tubing	1.7"
OD Drill Bit	7.87"
Cement weight	1400 kg/m
Cement used	SpectraCem NP + 0.9% FL-5 + 0.8% LCB-1000
Total Well Depth	2959'
Formation	Granite Wash
Treatment interval	2677.5' - 2611.64'
Gross interval	19.92' / 12.8'
Open hole used	7.87"
Mud weight used	1100 - 1300 kg/m ³
Bottom hole temp	86°F / 30°C
Casing Collars	-
Fluid salinity	46000 ppm

RESULTS

Casing area volume (inside) 2959.0'	19.5 m ³ / 19051 l / 159.8 Bbl / 5036.6 gal
Tubing area volume on 2611.64'	2.28 m ³ / 2285 l / 19.2 Bbl / 604.0 gal
Tubing area volume on 2677.5'	2.34 m ³ / 2342 l / 19.6 Bbl / 619.3 gal
Tubing area volume (inside) 2611.64'	1.17 m ³ / 1166 l / 9.8 Bbl / 308.2 gal
Tubing area volume (inside) 2677.5'	1.20 m ³ / 1195 l / 10.0 Bbl / 315.9 gal
Fluid Volume (Well) for 2611.64'	8.14 m ³ / 8135 l / 68.2 Bbl / 2150.7 gal
Fluid Volume (Well) for 2677.5'	8.11 m ³ / 8107 l / 68.0 Bbl / 2143.3 gal
Average Volume (Well) 2644.57'	8.12 m ³ / 8121 l / 68.1 Bbl / 2147.0 gal
Pump performers	4.0 Bbl/min
Working pressure	37.9 MPa (~ 5500 psi)
Equipment speed	4.0 - 5.0 mm/min corrected during the operation
Full length of the equipment (before tubing)	11.2'
Marker length	4.1'
One section of tubing length	31.5'
Installation of the first interval	2677' (Elongation 0.47' or 5.68")
Cut 1	2677.5' - 2679.14' ~ 2 hours
Lift 1	3'
Cut 2	2674.5' - 2676.14' ~ 2 hours
Lift 2	3'
Cut 3	2671.5' - 2673.14' ~ 2 hours
Lift 3	14'
Cut 4	2657.5' - 2659.14' ~ 2 hours
Lift 4	3'
Cut 5	2654.5' - 2656.14' ~ 2 hours
Lift 5	3'
Cut 6	2651.5' - 2653.14' ~ 2 hours
Lift 6	3'



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Cut 7	2648.5' - 2649.14'	~ 2 hours
Lift 7	38.5'	
Cut 8	2610.0' - 2611.64'	~ 2 hours
Shale shaker (on top of cuttings tank)	1 pump (500-600 l/min)	
If not shaker; then 1 filter	(0.125 particle size)	2 pumps (combined 1 m ³ /min)
Low pressure tubing plus connections	98.4 ft (30 m)	
Standard BOP equipment	Frac valve + 8 stripper rubbers for frac valve	
1 high-pressure (triplex) pump	minimum 35 MPa & 1 m ³ /min	
1 blender	25-50 kg/ m ³	
High pressure iron tubing & reverse flow manifold	30 m (98.4 ft) including connections	
Frac van	1	
Non-soapy brine from nearby well	25 m ³ /min (25 l/min)	
Friction reducer additive for	20 m (65.6 ft) or 3 water	
Abrasive quartz sand (abrasive grit*) 10/35	20 tons (2 nozzles), 40 tons (4 nozzles)	
The distance between cuts	1.36'	
Total cuts	12.8 ft / 19.92 ft	~ 16 hours
The area excavated surface	~ 200 ft ² (2 nozzles), 400 ft ² (4 nozzles)	
Total working time (with flushing and lifting)	16 hours	
Elongation of tubing (~ 30 MPa)	0.47' / 5.68"	
Time to reach the sand to the nozzles	~ 3 min	
Time to reach the sand to the surface	~ 12.0 min	
Pressure loss in tubing	~ 2.0 MPa	
Pressure drop across the nozzles	~ 35.3 MPa	
Friction coefficient	~ 1.5	
Pup joints	0.6 m + 1.5 m (2' + 5')	
Volume of working fluid in the surface system	19.5 m ³ / 19051 l / 159.8 Bbl / 5036.6 gal	
Water storage tank volume approximately (minimum)	19.5 m ³ / 19051 l / 159.8 Bbl / 5036.6 gal	
Cutting tank volume approximately (minimum)	10.0 m ³ / 9840 l / 83 Bbl / 2601 gal	



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HYDRO-SLOTTING PERFORATION PROGRAM

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Program Name
Well Name
Area Name
Date
Prepared by
Prepared for
Confidentiality notice:

MAXXWELL PRODUCTION INC. HYDRO-SLOTTING PERFORATION TECHNOLOGY PROGRAM

142090023639

Granite Wash - 142090023639

29-Jan-14

ANATOLI NIKOULINE - SLOTTING PERFORATION TECHNOLOGY ENGINEER (MAXXWELL PRODUCTION INC.)

FOR ADMIRALTY OILS LTD. (SASKATCHEWAN) (ADMIRALTY PINTO 02-01-002-05W2 (84G372 LEASE))

This Program has been furnished by ANATOLI NIKOULINE - HYDRO-SLOTTING PERFORATION TECHNOLOGY ENGINEER

for 142090023639

under the terms of a Service Agreement dated

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or any of 142090023639

MAXXWELL PRODUCTION INC.

associates.

WELL & COMPLETION DATA

Casing

(Fill in one dimensions cell only, inches or millimeters)

(Links for casings)

OD Casing	4.50	inches	mm	Drift	4		
ID Casing	4.00	inches	mm				
Other parameters		pound/ft	kg/m	Steel Grade	J-55 ST&C		

Tubing

(Fill in one dimensions cell only, inches or millimeters)

(Links for casings)

OD Tubing	2.38	inches	mm	Drift	1.7		
ID Tubing	1.70	inches	mm	Internal Yield			
Other parameters		pound/ft	kg/m	Steel Grade	J-55 EUE T&C		

Cement area

(Fill in one dimensions cell only, inches or millimeters)

(Links for casings)

OD Drill Bit	7.87	inches	mm	Mud Weight Used			
Cement weight		lb/gal	kg/m ³				
Cement used				Cement Grade			

Treatment

(Fill in one dimensions cell only, feet or meters)

Total Well Depth	2959	ft	m	TREATMENT INTERVALS: 2677.5' - 2611.64'			
Formation				Granite Wash			
Treatment interval	2611.64	ft to	2677.50	ft	or (l)	m to	m
Gross interval	65.86	ft	0.00	m	Intervals between the cutting 50 cm		

Additional

(Fill in one dimensions cell only)

Open hole used	7.87	inches	mm				
Mud weight used		lb/gal		kg/m ³			
Bottom hole temp	86.00	F°	30.00	C°		F°	
Casing Collars		ft		m			
Fluid salinity	46000.00	ppm					

				(ft ³)	(m ³)	(liters)	gal (US)	bbl (US)	bbl (oil)	bbl (UK)	gal (UK)	
Casing area volume (inside)	2959	ft / or	901.90	m	672.78	19.05	19051	5036.6	159.8	119.8	116.4	4190.6
Tubing area volume on	2611.64	ft / or	796.03	m	80.69	2.28	2285	604.0	19.2	14.4	14.0	502.6
Tubing area volume on	2677.5	ft / or	816.10	m	82.72	2.34	2342	619.3	19.6	14.7	14.3	515.2
Tubing area volume (inside)	2611.64	ft / or	796.03	m	41.17	1.17	1166	308.2	9.8	7.3	7.1	256.4
Tubing area volume (inside)	2677.5	ft / or	816.10	m	42.20	1.20	1195	315.9	10.0	7.5	7.3	262.9
Fluid Volume (Well) for	2611.64	ft / or	796.03	m	287.29	8.14	8135	2150.7	68.2	51.2	49.7	1789.5
Fluid Volume (Well) for	2677.5	ft / or	816.10	m	288.30	8.11	8107	2143.3	68.0	51.0	49.5	1783.3
Average Volume (Well)	2644.57	ft / or	806.06	m	288.80	8.12	8121	2147.0	68.1	51.1	49.6	1786.4
Volume of working fluid in the surface system and tank volume (min)					574	16	16242	4294	136	102	99	3573

SURFACE EQUIPMENT

The following surface equipment is required on site for hydro-slotting perforation technology completion:

RIG

		(ft ³)	(m ³)	(liters)	gal (us)	bbl (us)	bbl (oil)	bbl (UK)	gal (UK)
1	Rig, plus tubing trailer with tubing, to be spotted opposite rig.	574	16	16242	4294	136	102	99	3573
2	Water storage tank volume approximately (minimum)	347	10	9840	2601	83	62	60	2164
3	Cuttings tank volume approximately (minimum)								
4	Shale shaker (on top of cuttings tank); 1 pump (500-600 L/min) or 17.5-21.2 ft ³ /min or 132-159 gal (us) or 4.2-5.0 bbl (us).								
5	If not shaker, then 1 filter (0.125 particle size), 2 pumps (combined 1 m ³ /min) or 35.3 ft ³ /min or 264 (gal (US)/min or 8.4 bbl (US)/min								
6	30 m (98.4 ft) of low pressure tubing plus connections								
7	Standard BOP equipment – frac valve + 8 stripper rubbers for frac valve								

FRAC SERVICE

1	1 high-pressure (triplex) pump (minimum 35-41 MPa (5000-6000 psi) & 1 m ³ /min) or 35.3 ft ³ /min or 264 (gal (US)/min or 8.4 bbl (US)/min
2	1 blender (25-50 kg/m ³)
3	30 m (98.4 ft) of high pressure iron tubing, including connections & reverse flow manifold
4	Frac van

CONSUMABLES

1	20 m ³ /min (20 liters/min or 706.3 ft ³ or 5283 gal (us) or 168 bbl (us) lease water – non-soapy brine from nearby well (look in the "Rig" table)
2	Friction reducer additive for 20 m (65.6 ft) or 3 water
3	20 tons abrasive grit* (2 nozzles) or 40 tons (4 nozzles) Program must be finalized to determine quantity. (See the next Table)
4	1 sand hopper or dump truck compatible with blender

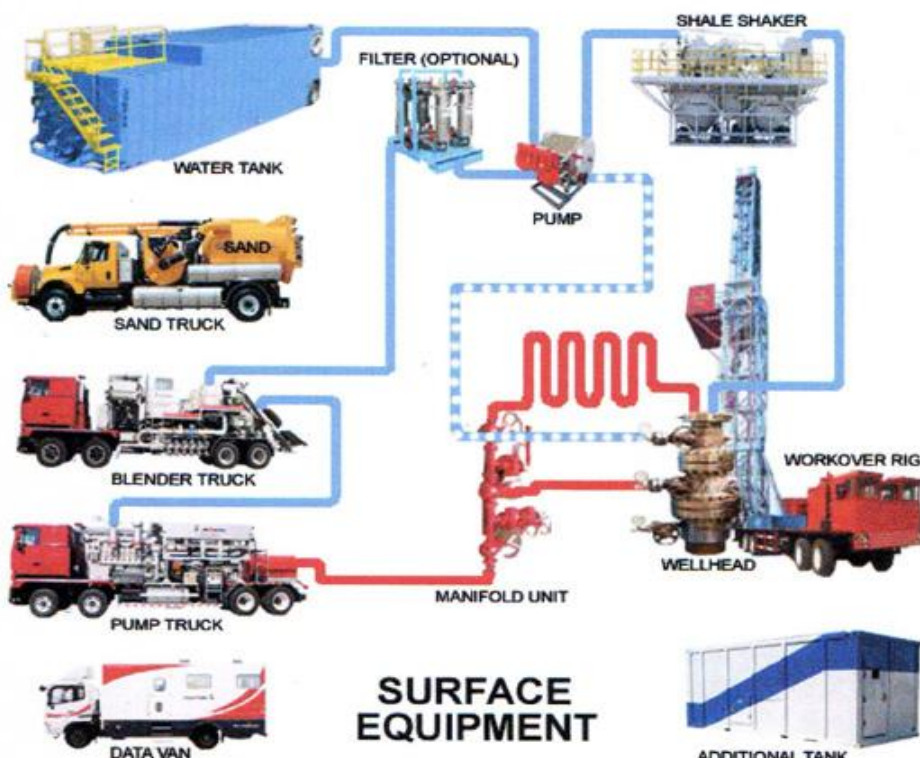


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CALCULATION THE ELONGATION OF THE TUBING UNDER THE WORKING FLUID PRESSURE, FIRST INTERVAL AND SECTIONS

A - engine length before nozzles
11.20 ft cm
B - collar with seat length
1.00 ft cm
C - additional pop joints section length
ft cm

or

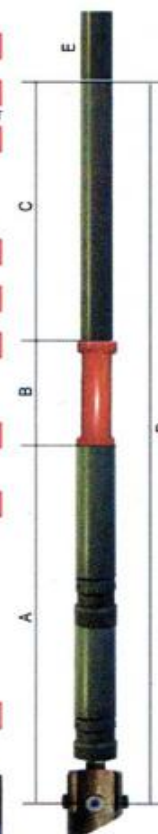
D - full length before tubing
11.20 ft cm
E - marker length
4.10 ft cm
F - one section of tubing length
31.50 ft cm

Pump performs (bbl/min)
bbl/min gal/min ft³/min
4.0
Wellhead pressure (psi)
psi Mpa atm
5000.0 35.0 352.0

Elongation of the tubing under the working fluid pressure :
from to
14.1 14.4 cm
0.46 0.47 ft
5.54 5.68 inch

Installation of the first interval :
ft m
2677.00

Number of sections (w/elong.):
Start Finish Difference
m m m
1597.50 1586.20 8.70
Sections: 85 tubings
Pup joints : 0.6m (2ft) 5
Pup joints : 1.5m (5ft) 4



1	Cut 1	from	2677.5	to	2679.14	ft			m
2	Cut 2	from	2674.5	to	2676.14	ft			m
3	Cut 3	from	2671.5	to	2673.14	ft			m
4	Cut 4	from	2657.5	to	2659.14	ft			m
5	Cut 5	from	2654.5	to	2656.14	ft			m
6	Cut 6	from	2651.5	to	2653.14	ft			m
7	Cut 7	from	2648.5	to	2649.14	ft			m
8	Cut 8	from	2610	to	2611.64	ft			m
Time for sand : 2.51 min 816.10 m 2.44 min 796.03 m Average 2.47 min Engine 4.0									

HYDRO-SLOTTING PROCESS

SEQUENCE OF PROCEDURES

INTERVAL 1

1	Put the "tubing" ball in-to the tubing, increase the pressure	6500	psi	45	Mpa	test the pressure during 1 min
2	Pressure-tested back line pressure of	1500	psi	10.5	Mpa	
3	Catch the "tubing" ball with back flushing.					
4	Increase the water pressure up-to	5000	psi	or 35	Mpa	and test tis mode during 1 min
5	Reduce the pressure and carefully start the sand	3	kg/m ³	See in the Converter		and slowly increase working fluid pressure
	up-to operat. mode	5000	psi	or 35	Mpa	
	Time to reach the oper. mode	1				
	Start cutting time = Pressure + Sand =	6.3	min	Cuting time for 1 interval =	100	min
					1.8	hours
6	After	100	min	or 1.8	hours	Cut the sand (I)
7	After	6.3	min	plus 10 min	wash	Cut the pressure (I) or reduce the pressure up-to 150 psi or 1 Mpa

If possible, keep the circulation. If the transition to the next interval takes considerable time, start the back flushing.

start the reverse circulation. When the flow is restored, to continue work. If circulation flow cannot be restored switch to direct flow, lower the tool in the well and replace the slurry in the well by clean formation water. Circulate the well until no sand or debris particles are present in the flow. Catch the ball on the shaker and lift the tool from the well.

If, to maintain the pressure, it is necessary to maintain the flow, we need to stop introduction of sand and start flushing with the clean formation water, then reverse the circulation, lower the down-hole tool and replace the slurry in the well by clean formation water. Circulate the well until no sand or debris particles are present in the flow. Catch the ball on the shaker and lift the tool from the well. The content of sand in the pulp cycle must be uniform and vary between of volume.

Stops are not permitted !



MAXXWELL PRODUCTION INC.

OIL WELL SLOTTING PROFESSIONAL SERVICE

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Necessary services, equipment and materials

1. Preparing the site. Inspection driveways, clean-up of the territory, coating gravel (if necessary).
2. Headwell installation.
3. Rig installation.
4. Lifting the service tubing from the well.
5. Preparing new tubing for the work.
6. Surface piping.
7. Tank with formation (layer) water (**19.5 m³ / 19051 l / 159.8 Bbl / 5036.6 gal**), the well should be filled with water too.
8. Should be able to heat water. The temperature of the working fluid in a closed circulation system must be around **86°F / 30°C**.
9. Empty tank of the same volume.
10. Shaker (for reset the waste abrasive quartz sand) **10.0 m³ / 9840 l / 83 Bbl / 2601 gal**.
11. Electricity and lighting on the site, safety, toilet, sitting area, meal.
12. Organization of garbage collection and disposal.
13. Abrasive quartz sand **20 tons (2 nozzles), 40 tons (4 nozzles)** in the special track.
14. Blender for controlled and uniform supply of sand.
15. Frac service (pump service) with frac-van (monitoring centre).
16. High pressure line and Manifold block.
17. Wireline service (for backsight logging).
18. Chemical treatment service (possible to use 5% HCL during slotting perforation or up-to 15-20% HCL after slotting perforation).

Rig service ~ **7** days.

Wireline service ~ **4** hours.

Pump service ~ **16** hours.

Requirements for the pump service

The main requirements for pumping equipment - stable work at various modes (stable pressure of the working fluid with minimal deviations, no jumps (pressure line on the monitor should be smooth). Supply of sand should be uniform (no batches). Pumping equipment should be easy to change and keep the pressure on specified modes up to tens psi, concentration of sand can support to the tenth proportions.

Working pressure (perforator with **2** or **4** nozzles):

psi		atm		MPa		bar	
4000	6000	272	408	27.5	42.0	27.5	420

Density (perforator with **2** or **4** nozzles):

g/l		Kg/m ²		Lb/gal		ppg	
20	60	20	50	0.017	0.038	8.42	8.57

Pumping speed (slurry rate) (perforator with **2** nozzles)

Bbl/min		l/min		m ³ /min		Gal/min	
2.5	5.5	298	775	0.3	0.7	79	173

Pumping speed (slurry rate) (perforator with **4** nozzles)



OIL AND GAS PRODUCT INTENSIFICATION AND RETRIEVAL TECHNOLOGY

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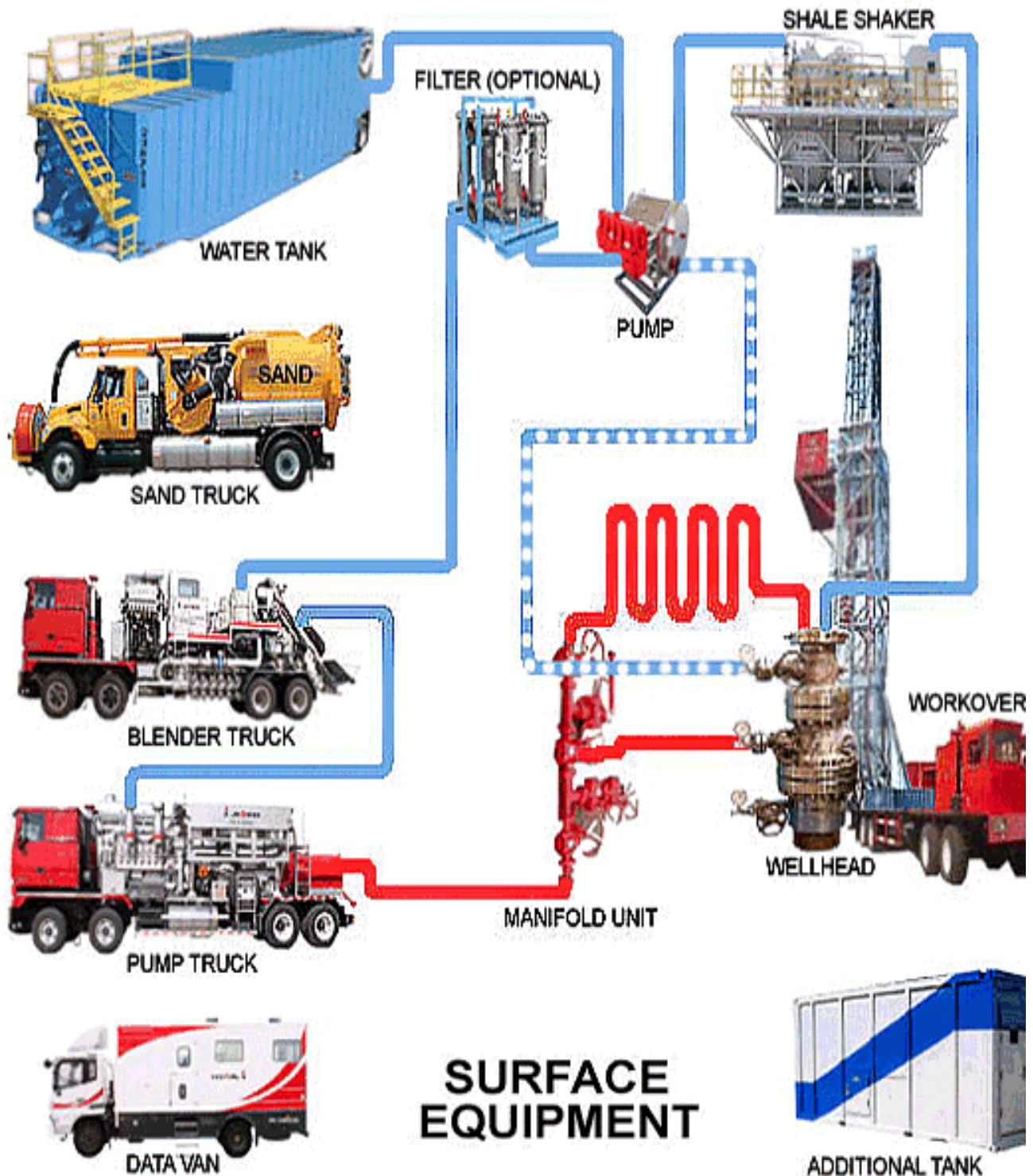
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Bbl/min		l/min		m³/min		Gal/min	
5	17	596	1800	0.6	2.0	270	900

The initial internal diameter of our nozzles 4.1 mm and 5.1 mm, become blurred to 8-9 mm.

General schematic of the surface equipment:





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Requirements for abrasive quartz sand:

For normal passage through the nozzles the size of sand shall not exceed **3.5 mm**. Best size of abrasive quartz sand **10/35**. If there is no such size, it is possible to use **20/40**.

PREPARATION FOR SLOTTING PERFORATION

PREPARATION

- a. Prepare site. MIRU. Spot rig, spot tubing trailer opposite rig if derrick cannot stand up tubing, cuttings tank, water storage tank, other equipment according to Surface Equipment schematic. Set up surface flow lines to be able to reverse flow csg/tbg (Manifold for csg/tbg reverse flow). Vacuum trucks bring **19.5 m³ / 19051 l / 159.8 Bbl / 5036.6 gal** brine and fill water storage tank. Record well pressure before taking off gauge. Bleed off any pressure (**< 0.7 MPa**). ND tree & NU Frac Valve. Pressure test Frac Valve. Kill well.
- ▶ During the process of hydro-slotting perforation it is recommended heating the water in the system (possibly steam generator). The temperature inside the well is **86°F / 30°C**. For prevent the cooling of the temperature during the process the water gets into the well from the surface, recommended heating water on the surface before the temperature of the working fluid in the pipes **86°F / 30°C**.
- ▶ Measure the length of the equipment assembly. During the lowering of equipment into the well to measure sections of tubing. Lower equipment to the well so that the perforator's nozzle at a depth **2677'** (Elongation **0.47'** or **5.68"**).
- b. (Stage 1 in the Tab) Make up and tally hydro-slotting equipment downhole assembly (equipment + transition adapter + test sub (marker) (~ **1 x 1.25 m**) + pup joints (~ **2 x 2 m, 1 x 1 m**). **RIH** with equipment and **PU** (~ **1 m**) tubing from trailer & drift & tally in hole. **RIH** with pup joints (~ **2 x 2 m, 1 x 1 m**): **0.6 m + 1.5 m (2' + 5')**.
- c. (Stage 2 in the Tab) Optional **RU** wireline loggers. **RIH** with **125.0 mm** gauge ring. If casing is new and no deformation is expected, and no fill is expected, well ID does not need to be.
- d. Warm up. **RU** frac equipment (pumps, blender, hopper, high-pressure lines, manifold, frac van). **RU** sand truck and conveyor belt to blender. Record water temperature and viscosity. Record **SI** pressure of tbg/csg. Test stripper rubber. Test surface lines.
- ▶ During the process of hydro-slotting perforation it is recommended heating the water in the system (possibly steam generator). The temperature inside the well is **86°F / 30°C**. For prevent the cooling of the temperature during the process the water gets into the well from the surface, recommended heating water on the surface before the temperature of the working fluid in the pipes **86°F / 30°C**.
- e. Conduct operational and safety meeting with all personnel on location, detailing all procedures and pressure limitations.
- ▶ Hydro-slotting operations. Refer to hydro-slotting procedures (next section). Program will be comprised of **33** main stages and includes the following vertical intervals, in order of excavation (from deepest to shallowest) (to be varied to avoid casing collars).
- ▶ Abrasive quartz sand (abrasive grit*) **10/35: 20 tons (2 nozzles), 40 tons (4 nozzles)**
- ▶ During the process can be adjusted the final intervals (depth).
- ▶ Take into consideration the elongation of tubing ~ **0.47'** or **5.68"**.
- ▶ The content of sand in the pulp cycle must be uniform and vary between **5% - 7%** of volume.
- ▶ If the pressure increase over **41.5 MPa**, accompanied by drop the tempo of pumping down, it is necessary to stop supply the sand, and turn reverse flushing.
- ▶ If the rate resumption - continue the process.
- ▶ If the circulation rate fails, turn reverse flushing.



► If the operating pressure drop below **10 Mpa** the equipment automatic charging to its original start position (interval).

► Lowering up the tool

► Cutting process

Installation of the first interval	2677' (Elongation 0.47' or 5.68")
Cut 1	2677.5' - 2679.14' ~ 2 hours
Lift 1	3'
Cut 2	2674.5' - 2676.14' ~ 2 hours
Lift 2	3'
Cut 3	2671.5' - 2673.14' ~ 2 hours
Lift 3	14'
Cut 4	2657.5' - 2659.14' ~ 2 hours
Lift 4	3'
Cut 5	2654.5' - 2656.14' ~ 2 hours
Lift 5	3'
Cut 6	2651.5' - 2653.14' ~ 2 hours
Lift 6	3'
Cut 7	2648.5' - 2649.14' ~ 2 hours
Lift 7	38.5'
Cut 8	2610.0' - 2611.64' ~ 2 hours

ONE DAY OPERATION

CORRELATION

stage	operation (procedure)	the depth (level)	distance	approximate time
1	Lowering the tool for 1 interval	2677' Elongation ~ 0.47' or 5.68"		~ 1 hour 30 min
2	Correlation log			~ 1 hour 25 min
3	Correction the installation	2677' Elongation ~ 0.47' or 5.68"		~ 0 hour 05 min

TIME : ~ **3** hour **00** min

SECOND (MAIN) DAY OPERATION

TUBING TESTING

stage	operation (procedure)	the depth (level)	distance	approximate time
4	Drop the testing ball			~ 0 hour 10 min
5	Tubing pressure testing	(within 1 min)		~ 0 hour 05 min
6	Reverse flushing			~ 0 hour 15 min
7	Catching the testing ball	(within 10 min)		~ 0 hour 05 min
8	Drop the valve (cutting) ball			~ 0 hour 15 min

TIME : ~ **0** hour **50** min

SLOTTING PERFORATION PROCESS

stage	operation (procedure)	the depth (level)	distance	approximate time
9	Cutting 1 interval (cutting down)	2677.5' ► 2679.14'	1.64'	~ 1 hour 45 min
10	Direct flushing			~ 0 hour 20 min
11	Lifting UP the tool for the 2 interval	2677.5' ► 2674.5'	3.0'	~ 0 hour 05 min
12	Cutting 2 interval (cutting down)	2674.5' ► 2676.14'	1.64'	~ 1 hour 45 min



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13	Direct flushing			~ 0 hour 20 min
14	Lifting UP the tool for the 3 interval	2674.5' ► 2671.5'	3.0'	~ 0 hour 05 min
15	Cutting 3 interval (cutting down)	2671.5' ► 2673.14'	1.64'	~ 1 hour 45 min
16	Direct flushing			~ 0 hour 20 min
17	Lifting UP the tool for the 4 interval	2671.5' ► 2657.5'	14.0'	~ 0 hour 15 min
18	Cutting 4 interval (cutting down)	2657.5' ► 2659.14'	1.64'	~ 1 hour 45 min
19	Direct flushing			~ 0 hour 20 min
20	Lifting UP the tool for the 5 interval	2657.5' ► 2654.5'	3.0'	~ 0 hour 05 min
21	Cutting 5 interval (cutting down)	2654.5' ► 2656.14'	1.64'	~ 1 hour 45 min
22	Direct flushing			~ 0 hour 20 min
23	Lifting UP the tool for the 6 interval	2654.5' ► 2651.5'	3.0'	~ 0 hour 10 min
24	Cutting 6 interval (cutting down)	2651.5' ► 2653.14'	1.64'	~ 1 hour 45 min
25	Direct flushing			~ 0 hour 20 min
26	Lifting UP the tool for the 7 interval	2651.5' ► 2648.5'	3.0'	~ 0 hour 15 min
27	Cutting 7 interval (cutting down)	2648.5' ► 2649.14'	1.64'	~ 1 hour 45 min
28	Direct flushing			~ 0 hour 20 min
29	Lifting UP the tool for the 8 interval	2648.5' ► 2610.0'	38.5'	~ 0 hour 30 min
30	Cutting 8 interval (cutting down)	2610.0' ► 2611.64'	1.64'	~ 1 hour 45 min
31	Direct flushing			~ 0 hour 20 min
32	Direct flushing with lowering the tool	2610.0' ► 2679.14'	69.14'	~ 1 hour 30 min
33	Lifting UP the tool to the surface			~ 1 hour 30 min
34	Disassembly the tool			~ 1 hour 30 min

TIME : ~ 22 hour 30 min

TOTAL TIME FOR THIS DAY : ~ 22 hour 30 min

- Lowering up the tool
- Cutting process

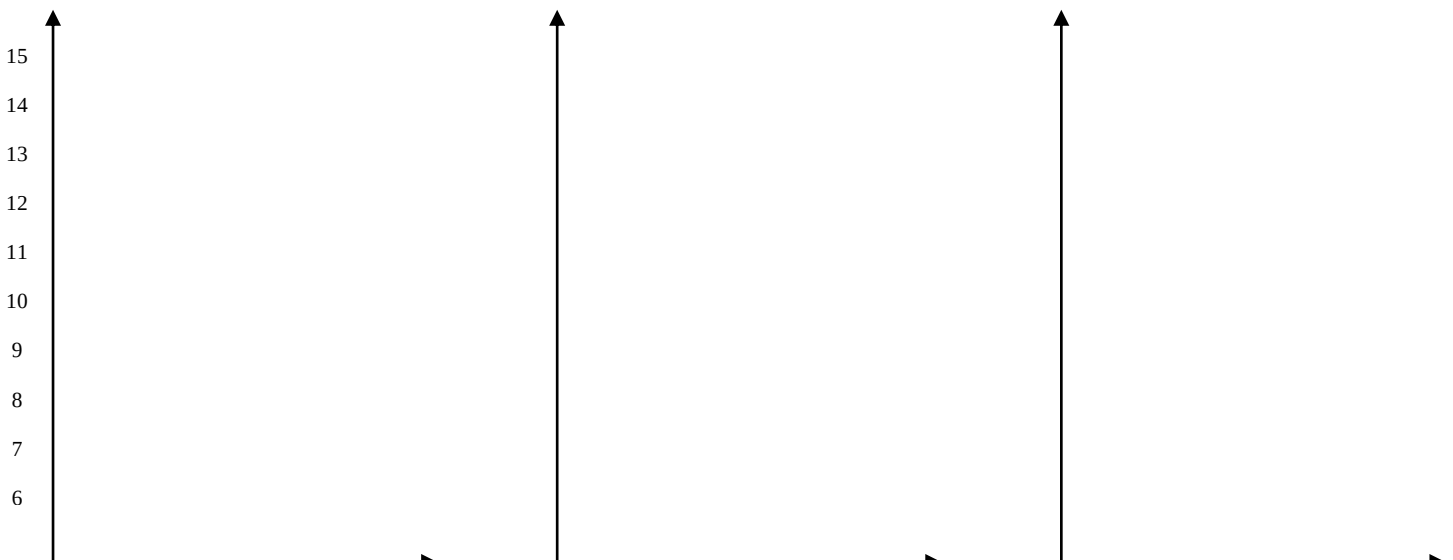
* During the process will need to make one round-trip operation (lifting the tool on the surface) for replacement the nozzles or perforator. It takes about 3 hours, so it is advisable to divide the whole slotting perforation operation for two working days, but it increases the cost of the pump service and water heating.

FOR NOTES

LOSS PRESSURE

WORKING PRESSURE

WELLHEAD PRESSURE





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POSSIBLE CHEMICAL TREATMENT

Continue completion, possible chemical treatment, and test well.

The approximate composition:

Acid:	15% HCl
Surface-active substances:	0.3%-0.5% (water-repellent agent for for debit wells)
Corrosion inhibitor:	0.5%
Stabilizer:	0.1%
Moderator of the reaction rate:	0.1% (in US three ready types).

Solution volume is calculated from the drainage radius equal **5** feet from the wellbore.